Bipartite Graph Attention Autoencoders for Unsupervised Change Detection Using VHR Remote Sensing Images

Meng Jia, Cheng Zhang, Zhiqiang Zhao, and Lei Wang

***Abstract*— Detecting land cover change is an essential task in very-high-spatial-resolution (VHR) remote sensing applications. However, because VHR images can capture the details of ground objects, the scenes of VHR images are usually complex. For example, VHR images usually show distinct appearances or features of the same object, aroused by noise, climate conditions, imaging angles, etc. To address this issue, this article proposes a novel unsupervised approach named bipartite graph attention autoencoders (BGAAEs) for VHR image change detection. BGAAE, a further improved way of using dual convolutional autoencoders based on the architecture of image translation, equips the encoder layers with a graph attention mechanism (GAM). To generate an effective difference image, it consists of two additional loss terms: the domain correlation and semantic consistency losses, in addition to the reconstruction loss. The domain correlation loss is designed based on the encoder layers, aiming to enforce the spatial alignment of deep feature representations of the unchanged objects and mitigate the influence of pixel changes on the learning objective. The semantic consistency loss focuses on ensuring the semantic feature consistency of the bitemporal images after transcoding and allows for more flexible transformations. The experimental results on four VHR image datasets demonstrate the superiority of the proposed method.**

***Index Terms*— Autoencoders, change detection, graph attention, remote sensing images, semantic consistency, very-high-spatial-resolution (VHR) images.**

I. Introduction

DURING the past decades, a new generation of remote sensing devices has provided us with a unique opportunity to conduct monitoring on our planet. Very-high-spatial-resolution (VHR) remote sensing images, whose resolution is less than one meter, are acquired to record the dynamic changes in land cover in the long term or short term [1]–[3]. In this context, plenty of data have been available to analyze more sophisticated targets, e.g., buildings [4], [5]. Detecting changes in VHR images is a fundamental but challenging task that intends to perform automatic techniques to comprehensively understand the regions of change between bitemporal images. This refers to a classification process that identifies changes through the joint analysis of images obtained in the same geographical location but at different times. In particular, it has been widely used in land cover monitoring (e.g., urban planning and deforestation) and natural disaster assessment (e.g., earthquakes, floods, and forest fires).

The performance of change detection is highly affected by multiview imaging, sensor resolution, noise, weather, illumination, and other interference factors in multitemporal remote sensing images. The existing techniques are divided into supervised and unsupervised techniques. The supervised method aims to learn statistical distributions, classifiers, or other prior information based on manually labeled data to acquire accurate interpretations of the changed regions of complex scenes. Its overall performance heavily depends on the accuracy of the constructed model [6]. However, models learned with a limited amount of data usually cannot provide satisfactory performance, especially for complex and large-scale remote sensing images in practical applications. Additionally, constructing labeled data is labor-intensive and time-consuming, limiting the generalization of supervised change detection techniques. Accordingly, unsupervised methods have recently attracted the attention of most researchers and have been rapidly developed.

Unsupervised change detection techniques usually comprise three steps: 1) data preprocessing; 2) difference image generation; and 3) difference image analysis. The preprocessing stage is designed to allow two remote sensing images to have consistent comparabilities, where the images are registered, pansharpened, and radiometrically normalized. Then, to generate effective difference images, pixel-based or region-based difference image generation algorithms are proposed to acquire the representation of the real change information by comparing the multitemporal remote sensing images pixel-by-pixel or object-by-object, such as principal component analysis (PCA) [7] and slow feature analysis (SFA) [8]. Finally, the resultant changes can be identified by analyzing the difference images. Based on this prototype, traditional unsupervised change-detection techniques have been developed for a long-term effort from remote sensing images. Typical traditional unsuper-

Manuscript received 19 April 2022; revised 27 June 2022; accepted 5 July 2022. Date of publication 13 July 2022; date of current version 21 July 2022. This work was supported by the National Natural Science Foundation of China under Grant 61806163 and Grant 62176146. *(Corresponding author: Meng Jia.)*

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Digital Object Identifier 10.1109/TGRS.2022.3190504

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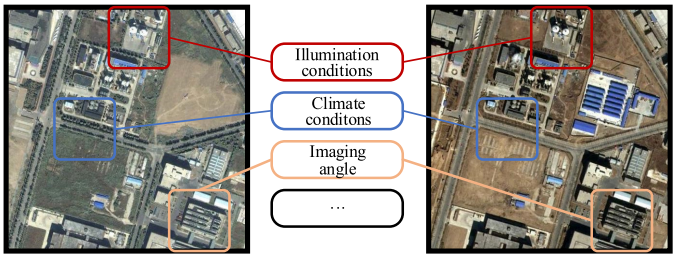


Fig. 1. Example of two images taken at different times. Different colored boxes correspond to different challenges: red represents different illuminations; blue represents different climates; and orange represents different imaging angles and so on.

vised methods were based on thresholding and clustering for change detection. However, the impact of noise and climatic conditions (e.g., seasons and illumination conditions) results in an inhomogeneous signature for multitemporal remote sensing images and makes the change detection task more difficult to address. Thus, the incorporation of spatial information can lead to a significant improvement to these traditional methods. Various methods are specifically designed for different scenarios [9]–[12]. Local neighborhood information has also been incorporated into the multiple kernel k-means clustering objective function to resist speckle for Synthetic Aperture Radar (SAR) change detection [9]. In [10], a nonlocal patch-based graph (INLPG) was proposed to detect changes, where the structures of bitemporal remote sensing images were compared considering modality-invariancy and resistance to various interference factors. In [11], contextual information among neighboring pixels was utilized to construct a Markov random field to detect changes. Jia and Wang [12] proposed constructing *a priori* by spatial characteristics and embedding it into the expectation-maximization (EM) iteration process. However, detecting the change has become more challenging due to the complex sensor characteristics, high resolution, and noise in imagery, as shown in Fig. 1. With the handcrafted features, the performance of the traditional models was limited despite its success in some applications.

Deep neural networks have recently received a wide range of attention due to their flexible feature learning ability in change detection for remote sensing images. Various deep learning networks have been actively studied for change detection, including convolutional neural networks (CNNs), deep belief networks (DBNs), generative adversarial networks (GANs), and deep autoencoder networks (DAENs) [13]–[17]. However, these deep neural networks tend to be trapped in a local minimum with inappropriate weights caused by the lack of training data, resulting in poor performance [17]. To train robust deep neural networks, a large amount of training data is needed. To address this issue, the intuitive and widely used strategy is unsupervised preclassification and sample selection followed by supervised network fine-tuning. Various traditional unsupervised change detection algorithms, such as joint Fuzzy C-means (JFCM) [15] and a hierarchical FCM [14], are utilized to generate training samples with pseudolabels to enrich the training samples. The dual-tree complex wavelet transform was introduced into CNNs to detect changes in SAR. To train the improved CNN model, pseudosamples were simulated using the existing data by a hierarchical FCM on the log-ratio image [14]. In [15], the DBN was trained based on the training images and their morphological features obtained via a JFCM classifier with a high probability of correct classification [15]. Niu *et al.* [13] proposed two steps of framework that consist of unsupervised feature learning and supervised fine-tuning, where the pseudolabels were obtained with the clustering method. Niu *et al.* [13] proposed adopting a conditional generative adversarial network (cGAN) to transform heterogeneous remote sensing images into some space, trained by pair samples selected via the generalized Kittler and Illingworth (GKI) thresholding method. Tang *et al.* [18] proposed an unsupervised model based on a graph convolutional network (GCN) and metric learning, in which a GCN was adopted to establish a contextual pattern with short and long ranges and metric learning was used to generate pseudodata.

Many works provide a promising potential of unsupervised models based on pseudodata generation. However, there are still limitations [19], which are as follows.

1) In real applications, acquiring adequate samples is one of the challenges in remote sensing image change detection to adapt to complex scenes and large-scale remote sensing images [2].
2) The final change detection accuracy highly depends on the quality of the training data, which is inevitably affected by the pseudolabel generation-based methods.

To overcome these limitations, another direction of research was conducted, where pseudolabel generation is not needed. Inspired by the principle of the deep kernelized autoencoder [20]–[22], code-aligned autoencoder (CAA) was proposed to circumnavigate the need for pseudolabels for unsupervised change detection in multimodal remote sensing images [23]. It obtains the change prior from pixel pair affinities at the input to align the code spaces, while reducing the influence of pixel variance. However, it is not straightforward to apply CAA to VHR images, which were originally designed for optical images. Therefore, it is necessary to investigate a method satisfying the spatial context without requiring a pseudolabel generation process.

In real applications, detecting the change in multitemporal VHR remote sensing images is highly affected by several factors, including noise, resolution, multiview imaging, weather, and illumination. Additionally, other interference factors make bitemporal VHR images show distinct appearances or features, and it is difficult to avoid detecting unchanged regions as changed ones. To address this issue, this article proposes image-to-image translation to extract semantic features for bitemporal image analysis, where the various objects and their composition describe the input semantics, avoiding interference factors on the performance of change detection. Image translation has been adopted in many applications. To study the sample transformation of face images to an analog sample of digital images, Taigman *et al.* [24] proposed a domain transfer network (DTN) that used a compound loss for encouraging the creation of samples to map samples from themselves. A dual adversarial autoencoder, Cross-GAN (XGAN), was proposed to jointly learn the domain-to-domain image translations in

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both directions while capturing shared semantic content across the two domains [25]. To translate between domains without paired input–output samples, Zhu *et al.* [26] introduced a cycle-consistent loss to enforce the translation to arrive back at the original domain.

Graph neural networks (GNNs) can generalize convolutional operations on arbitrary graph-structured data [27], [28]. It has shown the promising capability of performing a neural network-based aggregator that aggregates features from a node's local neighborhood [29]. This strategy can be used to detect changes in remote sensing images due to its applicability to unaligned bitemporal images. To overcome misalignment and illumination variations that degrade the detection accuracy, Yuan *et al.* [30] introduced a GNN-based framework, where 2-D and 3-D features were fused for accurate building change detection. A recent approach that began with graph attention networks (GATs) [31] applied attention mechanisms [32], [33] to graphs has become one extension of GNNs, which guarantees learning the significance between nodes and their neighbors and overcomes the absence of shift invariance and a lack of coordinate systems [34].

We propose an unsupervised change detection method based on the framework of CAA, which leverages a graph attention mechanism (GAM) and can ingest knowledge related to unchanged regions from an unlabeled image dataset. The contributions of the proposed method are summarized as follows.

1) We present a novel unsupervised change detection method, bipartite graph attention autoencoders (BGAAEs), which is based on the architecture of image translation. A GAM and semantic consistency of bitemporal VHR images are introduced into this architecture based on feature-level information. It should be pointed out that BGAAE is trained in a self-supervised learning manner without requiring training data.
2) We establish a domain correlation loss based on the GAM. On the one hand, due to its capability to work with spatially coupled representations between bitemporal images, GAM can effectively pass spatial information among the feature domains to enforce spatial alignment of deep feature representations of the unchanged objects. On the other hand, domain correlation establishes the relationship between the feature and image to enhance the representation of the consistent regions between the bitemporal images and mitigate the influence of changes on the learning objective.
3) Making use of the architecture of image translation, the semantic styles of the bitemporal images are preserved to ensure the image semantics' consistency. More specifically, instead of modeling relationships among any pairs of elements in pixel space, our BGAAE proposed a semantic consistency loss term to model the deep features of the input images after the encoder.
4) Intensive experiments on four datasets verify that the proposed model can effectively address complex VHR remote sensing change detection tasks. The implementation is available at https://github.com/xauter/BGAAE_CD.

The rest of this article is organized as follows. Section II reviews the related works. Section III describes the modeling and change detection process. To evaluate our method, experiments are designed in Section IV. Finally, the conclusion of this article is drawn in Section V.

II. Related Works

A. Difference Image for Optical Images

Difference image generation is a critical step in the change detection process. The quality of the difference image can clearly affect the whole performance of change detection. A high-quality difference image can highlight the changed regions and explicit the interference factors to distinguish the background regions. By comparing bitemporal images pixel-by-pixel or object-by-object, traditional image difference operators, such as image differencing [1], PCA [7], change vector analysis (CVA) [35], and INLPG [10], have been designed to provide real changed information about the original spectral behavior or extracted features. However, the scenes of optical images, especially VHR images, are complex and usually show distinct appearances or features of the same object. These methods rarely consider pseudochanges caused by noise, climate conditions, imaging angles, etc. The whole performance of change detection is usually limited by difference images obtained from the direct comparison of spectral values (or simple transformations).

Therefore, to effectively alleviate the influence of pseudochanges, deep neural network-based difference image operators have been designed by considering their strong and flexible feature learning ability. Liu *et al.* [37] proposed modeling the joint probabilistic model of bitemporal images by a bipartite CNN to identify the difference degree of each pixel. A symmetric convolutional coupling network (SCCN) [37] was proposed to transform the bitemporal images into a feature space, where the difference image can be calculated based on more consistent feature representations. By constructing CAAs [23], the bitemporal images can be reconstructed in their original domain and transferred to the opposite domain. Thus, the difference image can be conveniently generated by direct comparison.

B. Change Detection Networks With Attention Mechanism

The attention mechanism imitates human visual characteristics that selectively pay attention to a portion of information of interest while ignoring other low-correlation parts. An attention mechanism can be a kind of effective way to help a network enhance the feature representation of changes. By modeling context in spatial, temporal, and channel dimensions, attention-based change detection approaches can achieve better performance than purely convolution-based approaches [38].

To date, channel attention, spatial attention, and self-attention, which are effective in modeling context to increase the reception field of the model, have been applied to change detection. An attention module is designed based on information in global and local channel dimensions and

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embedded into a Siamese network to effectively extract, fuse, and select image features from different layers and branches [39]. Liu *et al.* [40] proposed a super-resolution-based change detection network (SRCDNet) with a stacked attention module (SAM). The SAM is integrated into the feature extractor to enhance the multiscale features [40]. To alleviate the influence of pseudochanges in the changed regions, Liu *et al.* [41] proposed imitating a visual attention mechanism to search the real changing region in difference image. Chen *et al.* [38] proposed enhancing high-level features of changes by using nonlocal self-attention to construct a compact set of semantic tokens. Due to the complexity of VHR images, a global coattention model was introduced into a pyramid feature-based attention-guided Siamese network to improve change detection results, which emphasizes the correlation between the input feature pairs [42].

In addition, several works have designed attention modules that composite spatial and channel information simultaneously to enhance feature extraction for capturing changes [43]–[45]. Chen *et al.* [46] proposed embedding dual attention modules into fully convolutional Siamese networks to obtain more discriminant feature representations of the changes. The latest efforts have focused on designing attention mechanisms to enhance the feature representations of changes. We borrow the idea of object tracking [47] and focus on describing the consistent regions (unchanged regions) between the bitemporal images for change detection, which realizes similarity learning via image feature cross correlation between bitemporal images.

III. Methodology

A. Framework of BGAAE

Given VHR images $X \in \mathbb{R}^{W\times H\times C}$ and $Y \in \mathbb{R}^{W\times H\times C}$ captured over the same geographical region at two dates t_1 and t_2, respectively. C is the number of channels. The proposed model detects the changes occurring between t_1 and t_2. However, due to various interference factors, such as illumination, climate, and imaging angle, objects with the same semantic concept usually exhibit a great difference in spectral characteristics between different periods [38], which can be considered to be in different styles. This problem is suitable for image-to-image translation: images are mapped into an intermediate domain to exclude the effect of interference factors while maintaining semantic consistency. To do so, it is a crucial task that learns a one-to-one mapping to the spectral characteristics relationship of the land cover between two different image domains. Therefore, based on the architecture of image translation [23], we develop a BGAAE to implement image transformation to map bitemporal VHR images. Consistent feature representations are captured by transforming the bitemporal VHR images X and Y into an intermediate domain $\mathcal{S}$ to detect the changes. Specifically, a set of pixelwise features represents the input image in the transformed intermediate domain $\mathcal{S}$. Mathematically, the two transformations are defined as follows:

$$S_x = F(X) \tag{1}$$

$$S_y = G(Y) \tag{2}$$

where $F(\cdot)$ and $G(\cdot)$ are the learned mapping functions that transform the bitemporal VHR image into the intermediate domain $\mathcal{S}$ and obtain S_x and S_y, respectively. Then, based on the feature representations in $\mathcal{S}$, the difference image can be conveniently calculated by the pixelwise distances between S_x and S_y as follows:

$$\Delta(i, j) = d\big(s_x(i, j), s_y(i, j)\big) \tag{3}$$

where Δ denotes the difference image, captured from the pixelwise difference between the characteristic representations of X and Y in $\mathcal{S}$. (i, j) represents the location of each pixel in the bitemporal images. $s_x(i, j)$ and $s_y(i, j)$ are the corresponding characteristic representations of X and Y at the same positions after transformation, i.e.,$S_x = \{s_x(i, j)|i = 1, \ldots, W, j = 1, \ldots, H\}$ and $S_y = \{s_y(i, j)|i = 1, \ldots, W, j = 1, \ldots, H\}$. $d(\cdot)$ is a pixelwise distance metric function, which is chosen according to the characteristics of the VHR image [19]. We prefer to use Euclidean distances for the limitation of the computational cost.

Specifically, we construct a change detection framework consisting of two autoencoders to implement $F(X)$ and $G(Y)$. Each autoencoder is associated with one of the two image domains x and y. Therefore, the BGAAE consists of two encoder–decoder subnetworks: 1) the encoder $E_x(X):X \rightarrow Z_x$ and the corresponding decoder $D_x(Z):Z_x \rightarrow \hat{X}$ and 2) the encoder $E_y(Y):Y \rightarrow Z_y$ and the corresponding decoder $D_y(Z):Z_y \rightarrow \hat{Y}$. Here, $\hat{X}$ and $\hat{Y}$ are the reconstructed outputs. Two encoder–decoder pairs are constructed by deep fully convolutional networks. Z is the code layer representation, and Z_x and Z_y specifically represent the feature representations in the code layers or the latent space domains of the encoders E_x and E_y, respectively. The weights are shared at the encoders of the two encoder–decoder subnetworks. The aim is to align the code layers of two fully convolutional autoencoders. Therefore, the bitemporal images can be transformed to a common latent space, where the feature pairs Z_x and Z_y for the bitemporal images can be generated. Then, the output of one encoder can be the input of both decoders. The bitemporal images can be reconstructed in their original domain and transferred to the intermediate domain $\mathcal{S}$ as well. Correspondingly, BGAAE is designed to be composed of three loss functions, leading the network to train to obtain the desired objectives. Fig. 2 gives the architecture of the proposed BGAAE method.

Under appropriate regularization, the bipartite autoencoders can be jointly trained to learn to map their inputs to the latent space domains. Then, the latent domain codes are projected into their original domain by the cascaded encoders and corresponding decoders to reconstruct them with high fidelity in output. In addition, data are mapped across by the opposite decoders, leading to sought style transformations. Then, (1) and (2) can be changed into

$$S_x = F(X) = D_y(Z_x) = D_y(E_x(X)) \tag{4}$$

$$S_y = G(Y) = D_x\big(Z_y\big) = D_x\big(E_y(Y)\big) \tag{5}$$

as depicted in Fig. 2. However, without any external forcing, the characteristic representations S_x and S_y in $\mathcal{S}$ will, in general, not be close, such that they cannot be directly

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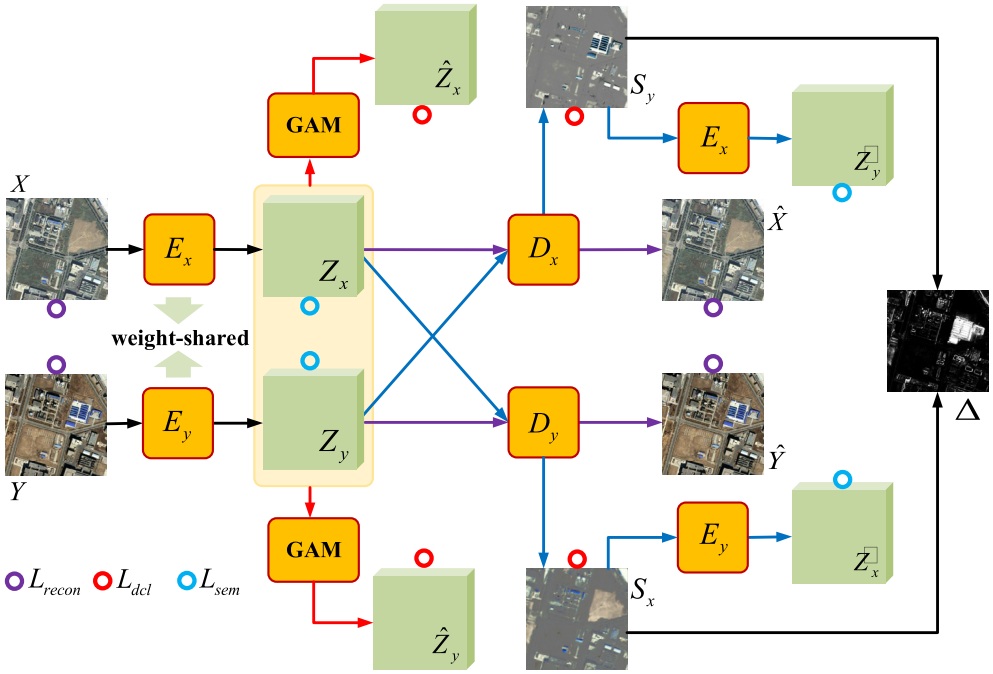


Fig. 2. Architecture of BGAAE consists of two encoder–decoder subnetworks with GAM. Data flows for different loss functions are marked by different colors: a reconstruction loss (purple), a domain correlation loss (red), and a semantic consistency loss (blue).

utilized to generate a difference image. Therefore, the loss terms are introduced to enforce their transformation alignment in the intermediate domain $\mathcal{S}$, both in the semantics and in the location of land covers. If the semantic features of S_x and S_y in $\mathcal{S}$ align successfully, the changes can be detected. The two mapping functions are implemented as bipartite autoencoders with deep fully convolutional networks. Thus, in our implementation, the image dimensions (W and H) can be retained throughout the hidden layers of the autoencoders to avoid bottlenecking [23].

The aim of BGAAE is to achieve reconstruction of the bitemporal images in their original domain with high fidelity and transformation in the intermediate domain. The bipartite autoencoder training can be carried out by minimizing a loss function concerning the parameter set ϑ of the constructed networks. The loss function $\mathcal{L}(\vartheta)$ consisting of three loss terms is defined to lead the network to train to obtain the desired objectives. Each loss function refers to a specific objective: a reconstruction loss function for bitemporal remote sensing image reconstruction, a domain correlation loss for exploring consistent regions, and a semantic consistency loss for alleviating the influence of interference factors.

B. Reconstruction Loss

Consider two training images X and Y of size $W \times H$ pixels. The first regularization for the bipartite autoencoders is to reconstruct the inputs with high fidelity in output. Here, we define the reconstruction loss term L_{recon}

$$\begin{aligned} L_{\text{recon}} &= d_2(X, \hat{X}) + d_2(Y, \hat{Y}) \\ &= d_2(X, D_x(E_x(X))) + d_2\big(Y, D_y\big(E_y(Y)\big)\big) \end{aligned} \tag{6}$$

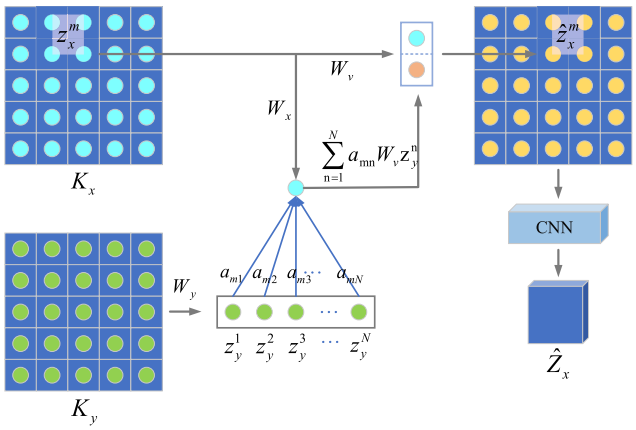


Fig. 3. Illustration of the GAM.

where $\hat{X}$ and $\hat{Y}$ are the reconstructed images from their input X and Y, respectively. d_2 is utilized to calculate the mean squared error (MSE) between the desired and predicted outputs.

C. Domain Correlation Loss

To align the image representation domain, we propose a novel domain correlation loss. Under the assumption that unchanged regions exhibit more spatial correlation [48], we explore the GAM in the latent space domain to estimate the spatial correlation between feature maps, without the need for manual calibration of parameters [37], and no additional prior knowledge [19], as shown in Fig. 3.

Given the bitemporal VHR images of X and Y, we first employ encoders E_x and E_y to map the inputs into the latent space domain, obtaining two feature representations

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Z_x and Z_y, respectively. Then, each $1 \times 1 \times C$ grid in Z_x and Z_y is considered to a node to generate a graph, where C is the number of feature channels. Let V_x denote a set of nodes including all nodes of Z_x, and let V_y correspondingly be the node set of Z_y. Inspired by Velikovi *et al.* [31], we construct a complete bipartite graph $\mathrm{K} = (V, E)$ to model the node-level relationships between the pre- and postchange images, where $V = V_x \cup V_y$ and $E = \{(u, v) \mid \forall u \in V_x, \forall v \in V_y\}$. We further define two subgraphs of K by $K_x = (V_x, \emptyset)$ and $K_y = (V_y, \emptyset)$.

The more similar feature between the pre- and postchange images, the more likely it is to be an unaltered region, and more spatial correction information should be conveyed to that region. For this reason, let e_{mn} represent the correlation between node $m \in V_x$ and node $n \in V_y$, which is defined as proportional to the similarity of the feature vectors of the two nodes. Here, the inner product of the feature vectors is used to calculate the similarity [49]. To adaptively capture a better representation between two nodes, linear transformations are applied to the node features. Therefore, for each $(m, n) \in E$, the correlation score can be calculated by

$$e_{mn} = \left(W_x z_x^m\right)^{\mathrm{T}} \left(W_y z_y^n\right) \tag{7}$$

where $z_x^m \in \mathbb{R}^C$ and $z_y^n \in \mathbb{R}^C$ represent feature vectors of node m and node n in the latent space domain. $W_x \in \mathbb{R}^C$ and $W_y \in \mathbb{R}^C$ are linear transformations applied to every node from V_x and V_y, respectively. In addition, they can be considered two weight matrices.

Based on the correlation score e_{mn}, an attention mechanism is performed on the nodes. We define a_{mn} to measure the influence of node n on node m, which is obtained by normalizing the correlation score e_{mn} using the Softmax function

$$a_{mn} = \frac{\exp(e_{mn})}{\sum_{k \in V_y} \exp(e_{ik})}. \tag{8}$$

That is, a_{mn} measures how much attention should be given to node m, according to the viewpoint of node n. Therefore, to propagate attention information from all nodes in V_y to the mth node in V_x, a linear transformation matrix $W_v \in \mathbb{R}^C$ is defined and applied to every node in V_y. Then, the mathematical formulation of an aggregated representation for node m becomes

$$v_m = \sum_{n \in V_y} a_{mn} W_v z_y^n. \tag{9}$$

As a consequence, a more powerful feature representation $\hat{z}_x^m$ is obtained by fusing the aggregated representation v_m with the node z_x^m

$$\hat{z}_x^m = \mathrm{ReLU}\left(v_m \parallel \left(W_v z_x^m\right)\right) \tag{10}$$

where $\parallel$ represents the vector concatenation operator. This attention mechanism is a single feed-forward neural layer. We compute all $\hat{z}_x^m$, $\forall m \in \mathrm{V}_x$ in parallel and then feed them into a convolutional layer to obtain an enhanced feature representation $\hat{Z}_x$. This strategy can make $\hat{Z}_x$ of the VHR image X enhanced by spatial information from the VHR image Y and explore the consistent regions between the bitemporal images as well.

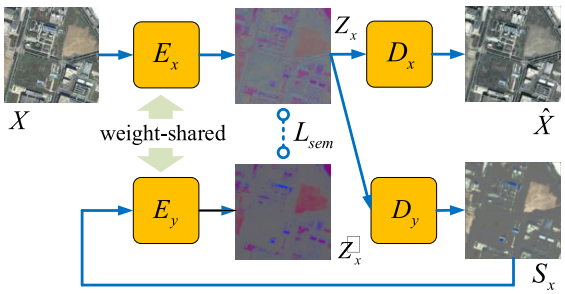


Fig. 4. Illustration of the sematic consistency loss of BGAAE for one of the bitemporal images.

Similarly, an enhanced feature map $\hat{Z}_y$ of VHR image Y can be calculated with respect to X following the method above.

Accordingly, the domain correlation loss L_{dcl} is computed as follows:

$$L_{\mathrm{dcl}} = d_2\left(S_x, \hat{Z}_x\right) + d_2\left(S_y, \hat{Z}_y\right) \tag{11}$$

where S_x and S_y are obtained by (4) and (5). Utilizing the domain correlation loss term L_{dcl}, we establish a node-to-node correspondence between the features and images. GAM is applied to calculate the domain correlation that can be regarded as a similarity between the bitemporal images. Therefore, the bipartite GAM can be considered to explore the consistent regions between the bitemporal images in the latent space domains and obtain the enhanced feature representations by using spatial information from the opposite images. More specifically, it takes the features from one image as kernels to directly perform convolution operations on the search region in another image and obtains enhanced features. In essence, the correlation can be regarded as a similarity calculation between the bitemporal images, and the obtained features are similarity maps for bitemporal images. Therefore, the bipartite GAM can be considered to obtain the consistent regions after the domain transformation.

D. Semantic Consistency Loss

Although objects in images X and Y usually exhibit greatly different spectral characteristics, these can be considered with the same semantic concept but in different styles. Mapping images into an intermediate domain $\mathcal{S}$ can effectively exclude the effect of interference factors. Semantic consistency implies that when images are transformed from one domain to another, deep features should maintain their relevance.

Inspired by the DTN [24], we introduce a semantic consistency loss term L_{sem}. The purpose is to let $F(\cdot)$ and $G(\cdot)$ learn features from each other to explore an intermediate domain $\mathcal{S}$. In particular, semantic consistency loss acts at the feature level, which is a further improvement of cycle consistency [26], allowing for more flexible transformations. Fig. 4 illustrates the sematic consistency loss of BGAAE for one of the bitemporal images. Therefore, to obtain the retransfomed feature representations $\tilde{Z}_x$ and $\tilde{Z}_y$, the transformations S_x and S_y are then encoded by

$$\tilde{Z}_x = E_y(S_x) \tag{12}$$

$$\tilde{Z}_y = E_x\left(S_y\right). \tag{13}$$

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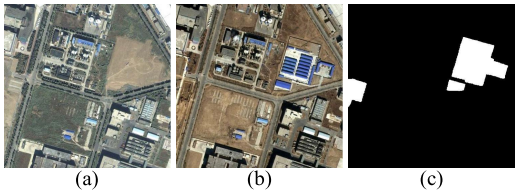


Fig. 5. Beijing dataset: (a) image acquired in 2012; (b) image acquired in 2013; and (c) ground truth image.

Thus, L_{sem} is defined in the latent space domain as

$$L_{\text{sem}} = d_2\big(Z_x, \tilde{Z}_x\big) + d_2\big(Z_y, \tilde{Z}_y\big) = d_2\big(E_x(X), E_y(F(X))\big) + d_2\big(E_y(Y), E_x(G(Y))\big) \quad (14)$$

where d_2 is the MSE defined above, and the encoders and the decoders come from (4) and (5). L_{sem} is critical for representing the error between image features. The aim of minimizing L_{sem} is to make the features of the original images X and Y close to those of the transcoded images S_x and S_y. More consistent feature representations in the code layers or the latent space domains can further generate ideal transcoded images in the intermediate domain $\mathcal{S}$ for accurate difference image generation.

E. Learning

The total loss is defined as the weighted average of the three loss terms, as follows:

$$L_{\text{total}} = L_{\text{recon}} + \alpha L_{\text{dcl}} + \beta L_{\text{sem}} \quad (15)$$

where the weights α and β are the hyperparameters to balance the three loss terms. After the training, the expected difference image Δ could be generated. Then, we chose Gaussian filtering [50] to regularize Δ, assuming that pixels are usually locally surrounded by pixels with the same labels. Consequently, the change map is detected by the classical Otsu method [51]. The key of this threshold selection algorithm is to obtain the optimal threshold, so all the pixel values are compared with the threshold. Greater than the threshold, we consider the pixels to be changed, and vice versa.

IV. Experimental Study

A. Experimental Settings and Datasets

To verify the performance of the proposed BGAAE, we conduct experiments on four public datasets, including the Beijing dataset [10], Nanjing University of Science and Technology (NJUST) dataset [36], Számítástechnikai és Automatizálási Kutatóintézet (Institute for Computer Science and Control) (SZADA) dataset [52], and Building Change Detection Dataset (BCDD) [40]. For the Beijing dataset, a pair of RGB VHR images in the Beijing area was collected from Google Earth on September 30, 2012, and March 4, 2013. The image size is $500 \times 500 \times 3$ with a spatial resolution of 1 m/pixel. The detection target is the changed buildings, as shown in Fig. 5.

The NJUST dataset [36] includes VHR images of the Nanjing University of Science and Technology, Nanjing, China, collected from Google Earth in September 2013 and March 2019. As shown in Fig. 6, the detection target is the changed buildings. The image size is 523×447, which covers an area of 0.59×0.5 km^2 with a spatial resolution of 1.2 m/pixel.



Fig. 6. NJUST dataset: (a) image acquired in 2013; (b) image acquired in 2019; and (c) ground truth image.



Fig. 7. SZADA dataset: (a) image acquired in 2000; (b) image acquired in 2005; and (c) ground truth image.

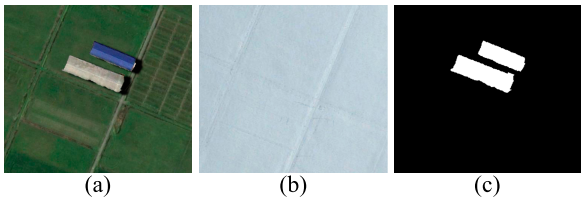


Fig. 8. BCDD dataset: (a) image acquired in 2012; (b) image acquired in 2016; and (c) ground truth image.

The SZADA dataset [52] contains seven pairs of VHR images taken by the Hungarian Institute of Geodesy Cartography and Remote Sensing (FÖMI), Budapest, Hungary, and Google Earth in 2000 and 2005. One pair (SZADA/7) (Fig. 7) is employed in this experiment. The image size is 500×500 with a spatial resolution of 1.5 m/pixel. Although Google Earth can offer abundant VHR images at different times, it often suffers interference factors (e.g., atmospheric conditions, climate, season, and imaging angle).

The BCDD [40] dataset includes bitemporal images obtained in February 2012 and April 2016. The whole dataset is $32\,507 \times 15\,354$ pixels in 20.5 km^2 with a spatial resolution of 0.2 m/pixel. As shown in Fig. 8, the detection target is the changed buildings. Since the images were acquired before and after the earthquake, the altered areas reflect the reconstruction of buildings. In this experiment, a 500×450 image pair is used, where changes are severely affected by season.

The proposed BGAAE was deployed with fully convolutional networks, where the encoder and decoder consist of three $3 \times 3 \times 50$ convolutional layers followed by LeakyReLU activation layers, one $3 \times 3 \times C$ convolutional layer, and a Tanh layer. The input size can be preserved by applying padding to the feature maps. The input bitemporal image data have been normalized between -1 and 1. A Tanh layer was used as the last decoding layer to rescale the data between -1 and 1, as this has been shown to speed up convergence

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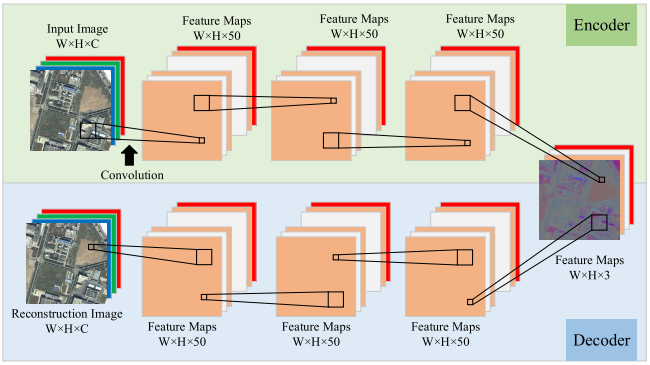


Fig. 9. Illustration of a pair of encoder–decoder structures in BGAAE.

as recommended in [53]. Fig. 9 gives a pair of encoder–decoder structures in BGAAE. BGAAE was implemented with TensorFlow Libraries and evaluated on a workstation with GeForce RTX 2080 Ti and 11G RAM. For a total number of 1000 training epochs, Adam is selected as the optimizer to facilitate model convergence. The learning rate is set to 1×10^{-3}.

B. Evaluation Criteria

Two criteria were utilized to evaluate the difference image, including the receiver operating characteristic (ROC) curve [54] and the area under the ROC curve (AUC). To evaluate the performance of the proposed BGAAE, the resultant change detection map is generated by the classical Otsu method. Then, four quantitative assessment criteria are adopted via comparison with the ground truth map: false positive (FP), false negative (FN), overall accuracy (OA), and Kappa coefficient (KC). These metrics can be calculated as follows for each binary change detection map:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{FP} + \mathrm{FN} + \mathrm{TP} + \mathrm{TN}} \tag{16}$$

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{17}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{FN} + \mathrm{TN}) \times (\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{18}$$

where TP and TN are the true positive and true negative, respectively.

C. Compared Methods

Qualitative and quantitative comparisons with eight popular unsupervised change detection methods are adopted as competitors to demonstrate the performance of BGAAE, including three traditional and five deep-learning models. They are summarized as follows.

1) PCA-$k$means[1] [7] clusters using k-means clustering PCA feature vectors extracted from nonoverlapping blocks of the absolute difference images.
2) CVA[1] [9] adopts the magnitude of the change vector in polar coordinates between the bitemporal images to detect changes.
3) Deep change vector analysis (DCVA)[2] [5] extracts spatial contextual information using a pretrained CNN, followed by DCVA to identify altered regions.
4) Deep kernel PCA convolutional mapping network (KPCA-Net)[3] [55] is an unsupervised deep Siamese kernel PCA convolutional mapping network.
5) DBN [15] consists of unsupervised feature learning and preclassification, followed by supervised fine-tuning with DBN.
6) A symmetric CNN (SCCN)[4] [37] utilizes an unsupervised deep convolutional coupling network to detect changes in remote sensing images.
7) CAAs[4] [23] adopt a dual fully convolutional network embedded into DAE.
8) INLPG[5] [10] models change by using the consistency of image structures of bitemporal remote sensing images.

[1]https://github.com/Bobholamovic/ChangeDetectionToolbox

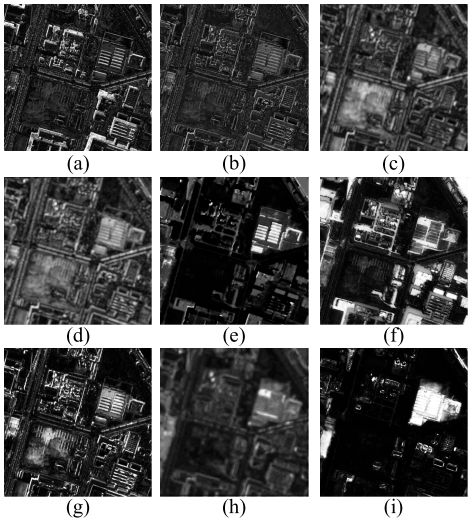


Fig. 10. Difference images obtained by different methods on the Beijing dataset: (a) PCA-$k$means; (b) CVA; (c) DCVA; (d) KPCA-Net; (e) DBN; (f) SCCN; (g) CAA; (h) INLPG; and (i) BGAAE.

D. Results on Beijing Dataset

The difficulty of this dataset lies in the pseudochanges mainly occurring in the shadows of buildings and the changes in grasslands caused by different imaging angles and climates. The difference images of this dataset generated by various change detection methods are shown in Fig. 10. By observing the difference image in Fig. 10, the shadow of buildings severely impacts the generated difference images, leading to many false detections of edges. BGAAE has a good inhibitory effect on pseudochanges, and it can most completely detect the changed regions. Correspondingly, the ROC curves of those

[2]https://github.com/sudipansaha/dcvaVHROptical
[3]https://github.com/I-Hope-Peace/KPCAMNet
[4]https://github.com/llu025/Heterogeneous_CD/tree/master/Code-Aligned_Autoen coders
[5]https://github.com/yulisun/INLPG

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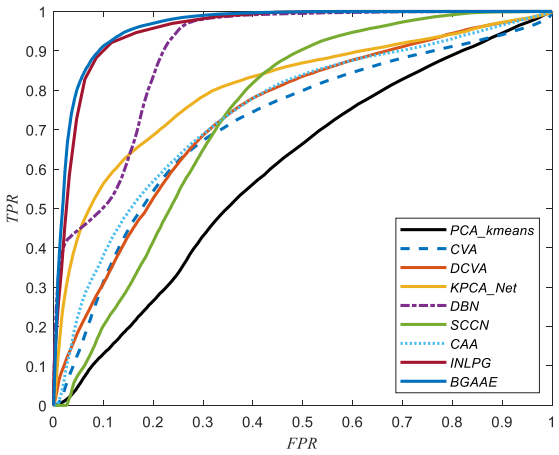


Fig. 11. ROC curves of difference images on the Beijing dataset.

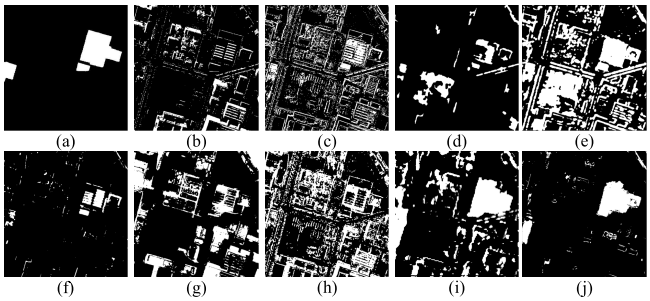


Fig. 12. Change maps obtained by different methods on the Beijing dataset: (a) ground truth; (b) PCA-*k*means; (c) CVA; (d) DCVA; (e) KPCA-Net; (f) DBN; (g) SCCN; (h) CAA; (i) INLPG; and (j) BGAAE.

difference images are illustrated in Fig. 11. It can be seen from the ROC curves that the difference image generated by BGAAE has the best performance.

Furthermore, we conducted the classical Otsu [50] on these difference images obtained by the abovementioned methods to generate the change map. Fig. 12 and Table I show the qualitative and quantitative results, respectively. As shown in Fig. 12(b), (c), and (i), the traditional methods provide poor detection accuracy for building change. In contrast, the deep learning models KPCA-Net, DCVA, SCCN, and CAA can effectively highlight the altered regions [Fig. 12(d)–(g)]. However, they cannot overcome the pseudochanges from building shadows well, leading to larger FPs. The change map of the DBN is not as good as expected for the loss of detail [see Fig. 12(f)], leading to a large FN. Because INLPG considers the relationship between image structures, it can detect change regions more completely and has the lowest FN. Nevertheless, boundary detection of changed regions is not clear enough in the change map of INLPG. As shown in Fig. 12(j), the proposed BGAAE precisely detects the altered regions due to the GAM and semantic consistency. As summarized in Table I, the proposed BGAAE performs the best in terms of AUC (96.33%), OA (94.83%), and KC (65.72%). Additionally, the improvements in KC are achieved by 62.85% (over PCA-*k*means), 39.72% (over DCVA), 53.73% (over SCCN), and 44.17% (over CAA).

TABLE I

RESULTS OF CHANGE DETECTION ON THE BEIJING DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
PCA_*k*means	0.5963	23879	16895	0.8369	0.0287
CVA	0.7152	43339	9396	0.7891	0.1851
DCVA	0.7380	14512	13084	0.8896	0.2600
KPCA_Net	0.8081	78721	3803	0.6699	0.1687
DBN	0.8979	**2742**	12690	0.9383	0.4428
SCCN	0.7394	47546	11198	0.7650	0.1199
CAA	0.7466	31185	10688	0.8325	0.2155
INLPG	0.9550	36917	**1263**	0.8473	0.4229
BGAAE	**0.9633**	7398	5517	**0.9483**	**0.6572**

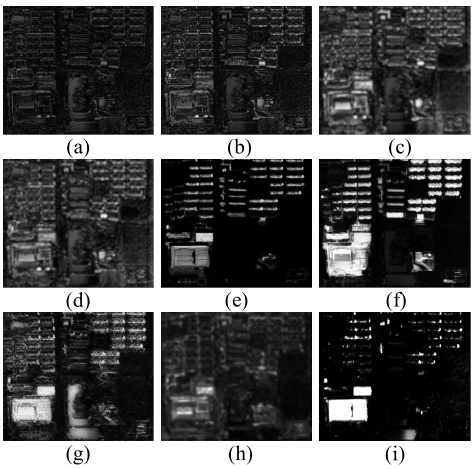


Fig. 13. Difference images obtained by different methods on the NJUST dataset: (a) PCA-*k*means; (b) CVA; (c) DCVA; (d) KPCA-Net; (e) DBN; (f) SCCN; (g) CAA; (h) INLPG; and (i) BGAAE.

Overall, the performance of BGAAE is the best among all these methods.

E. Results on the NJUST Dataset

In the NJUST dataset, the main change is caused by the renovation of the buildings. The pseudochange is primarily due to different light intensity conditions, such that the surface of the lake and the buildings show differences in spectral characteristics. Figs. 13 and 14 show the difference images and the corresponding ROC curves, respectively. As shown in Fig. 13(a) and (b), the difference images of traditional PCA-*k*means and CVA are greatly affected by the light intensity, and the change regions cannot be basically distinguished from the background. It can be seen from the ROC curves in Fig. 14 that these two traditional change detection methods may not be suitable for processing this dataset since their ROC curves are far from the top left corner. As shown in Fig. 13(c) and (e)–(g), DCVA, DBN, SCCN, and CAA can clearly depict the changed regions while effectively reducing false detections. However, the suppression effect on the background is relatively poor. In particular, because INLPG considers the structural information of images, it can achieve performance similar to deep learning technologies. In addition, Fig. 13(i) shows that the contrast between the altered and

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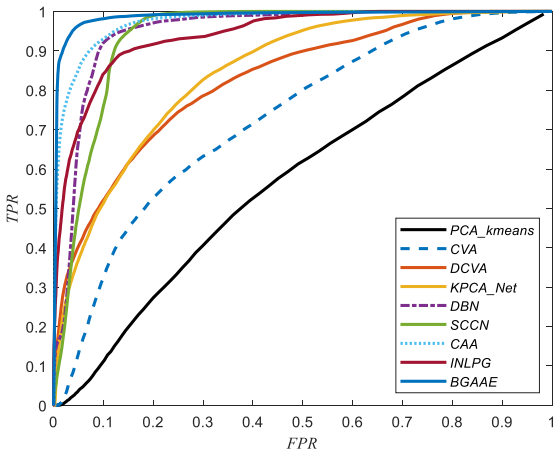


Fig. 14. ROC curves of difference images on the NJUST dataset.

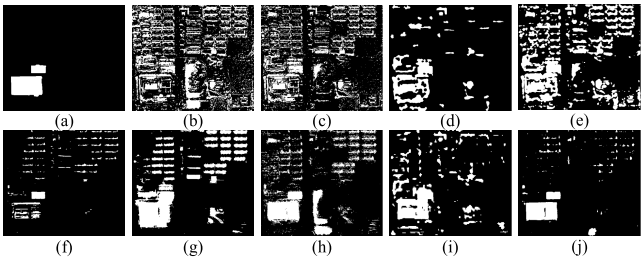


Fig. 15. Change maps obtained by different methods on the NJUST dataset: (a) ground truth; (b) PCA-*k*means; (c) CVA; (d) DCVA; (e) KPCA-Net; (f) DBN; (g) SCCN; (h) CAA; (i) INLPG; and (j) BGAAE.

TABLE II
RESULTS OF CHANGE DETECTION ON THE NJUST DATASET

Method	AUC	*FP*	*FN*	*OA*	*KC*
PCA_*k*means	0.5714	62064	7624	0.7019	0.0378
CVA	0.7290	42660	6090	0.7915	0.1369
DCVA	0.8265	18665	4281	0.9018	0.3745
KPCA_Net	0.8453	53305	3058	0.7589	0.1794
DBN	0.9449	6000	7948	0.9403	0.3693
SCCN	0.9367	33367	**635**	0.8546	0.3591
CAA	0.9741	26548	652	0.8837	0.4208
INLPG	0.9402	21817	2023	0.8980	0.4257
BGAAE	**0.9875**	**4534**	1160	**0.9756**	**0.7880**

unaltered regions is the most obvious due to the introduction of the GAM in the BGAAE.

The compared methods are qualitatively and quantitatively compared in Fig. 15 and Table II, respectively, showing the outperforming performance of the proposed BGAAE in terms of most criteria. For example, the enhancements in OA are 27.37% (over PCA-*k*means), 7.38% (over DCVA), 12.1% (over SCCN), and 9.19% (over CAA). However, the FN of our model is slightly higher than that of SCCN and CAA. The value of KC for BGAAE is greatly increased, up to 42.89% and 36.72% higher than that of SCCN and CAA, respectively.

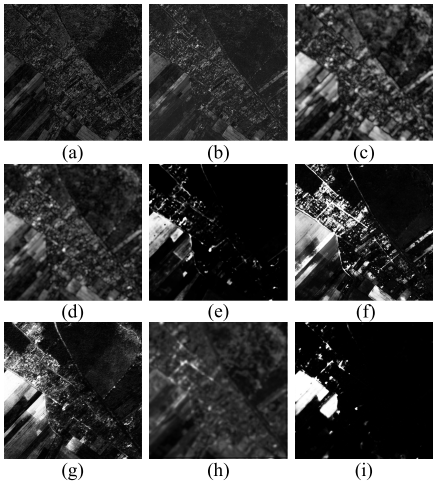


Fig. 16. Difference images obtained by different methods on the SZADA dataset: (a) PCA-*k*means; (b) CVA; (c) DCVA; (d) KPCA-Net; (e) DBN; (f) SCCN; (g) CAA; (h) INLPG; and (i) BGAAE.

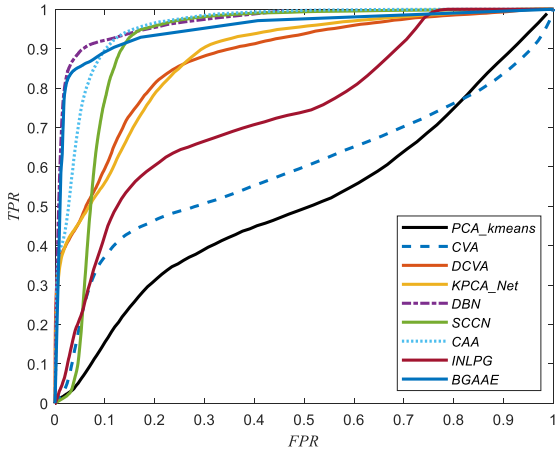


Fig. 17. ROC curves of difference images on the SZADA dataset.

F. Results on SZADA Dataset

The SZADA dataset includes various irregular changes, including constructed houses, new plowland, and roadway construction. The pseudochanges are composed of large forests, buildings, and plowlands with different spectral characteristics due to various atmospheric conditions and seasons. The qualitative and quantitative change detection results are depicted in Figs. 16–18 and Table III.

It can be seen from the qualitative results of difference images in Fig. 16 and their ROC curve in Fig. 17 that the performance of the deep learning related methods (i.e., DCVA, KPCA-Net, DBN, SCCN, CAA, and BGAAE) is better than that of the traditional methods (i.e., PCA-*k*means, CVA, and INLPG). Although almost all the comparative methods can highlight the changed regions, due to interference factors, some pseudochanges are included in these difference images. Fig. 18 shows all the resultant change detection maps. DCVA and SCCN tend to reveal all possible changes, so their change detection results are refined [see Fig. 18(d) and (g)]. In particular, we found that the performance of the DBN is close

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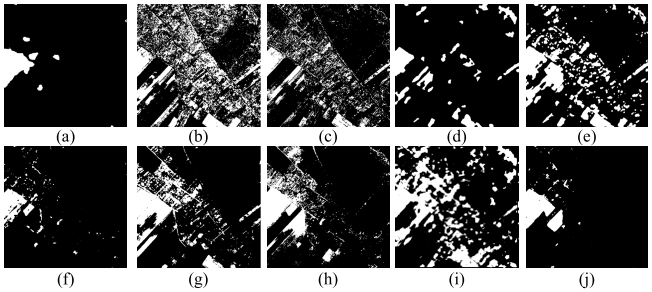


Fig. 18. Change maps obtained by different methods on the SZADA dataset: (a) ground truth; (b) PCA-*k*means; (c) CVA; (d) DCVA; (e) KPCA-Net; (f) DBN; (g) SCCN; (h) CAA; (i) INLPG; and (j) BGAAE.

TABLE III
RESULTS OF CHANGE DETECTION ON THE SZADA DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
PCA_*k*means	0.5112	78148	7190	0.6586	0.0289
CVA	0.5938	38230	7092	0.8187	0.1301
DCVA	0.8746	17521	7031	0.9018	0.2683
KPCA_Net	0.8769	43558	3104	0.8134	0.2284
DBN	**0.9683**	**3403**	3752	**0.9713**	0.6995
SCCN	0.9119	31778	1287	0.8677	0.3582
CAA	0.9559	26168	**1130**	0.8908	0.4145
INLPG	0.7404	59789	4527	0.7427	0.1282
BGAAE	0.9511	5671	2324	0.9680	**0.7054**

to that of the BGAAE. Fig. 18(f) and (j) shows that BGAAE is more robust to background interference (pseudochanges). Moreover, unlike other compared methods, BGAAE can detect the changed region while preventing detail loss, mainly due to the GAM and semantic consistency.

Table III summarizes the quantitative results, showing the outperforming performance of the proposed model. The improvements in the KC score are achieved by 67.65% (over PCA-*k*means), 57.53% (over CVA), 43.71% (over DCVA), 47.7% (over KPCA-Net), 0.59% (over DBN), 34.72% (over SCCN), 29.09% (over CAA), and 57.72% (over INLPG).

G. Results on the BCDD Dataset

For the BCDD dataset, the selected region is more severely affected by seasonal changes, and the main change comes from the demolition of two buildings. The pseudochanges are mainly affected by the heavy snow climate and different atmospheric conditions. The background has varied from dark green grass to snow-shrouded ground.

From the difference images generated by various methods in Fig. 19, we can see that the buildings can be well-separated from the background by these methods, but the smaller buildings are not detected by the DBN and INLPG methods. They falsely detected an area of plowland as a changed region. We can draw the same conclusion from the corresponding ROC curves in Fig. 20. Compared with the competitors, BGAAE detects all changed regions and suppresses pseudochanges. This is inseparable from the introduction of the GAM.

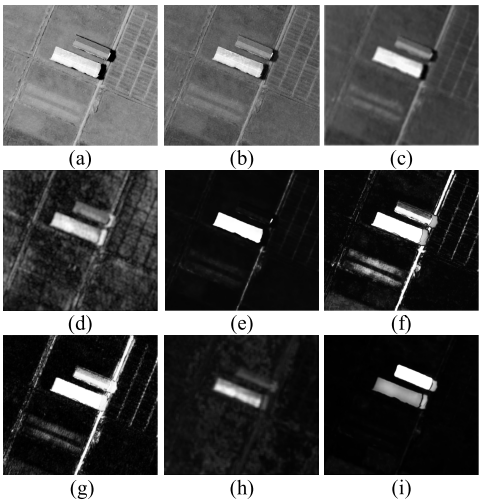


Fig. 19. Difference images obtained by different methods on the BCDD dataset: (a) PCA-*k*means; (b) CVA; (c) DCVA; (d) KPCA-Net; (e) DBN; (f) SCCN; (g) CAA; (h) INLPG; and (i) BGAAE.

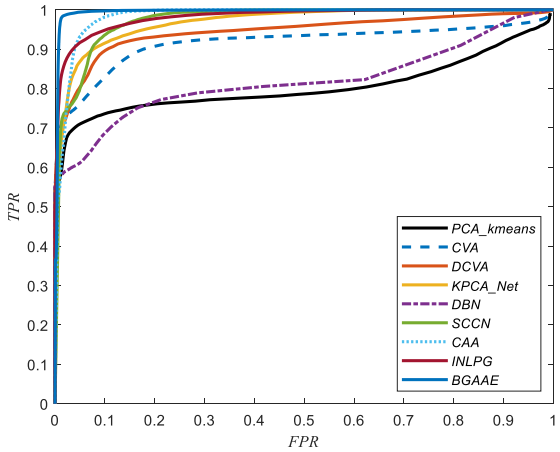


Fig. 20. ROC curves of difference images on the BCDD dataset.

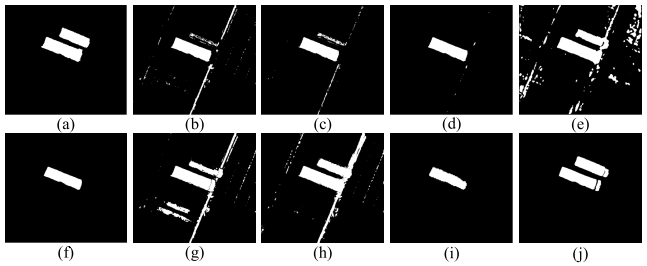


Fig. 21. Change maps obtained by different methods on the BCDD dataset: (a) ground truth; (b) PCA-*k*means; (c) CVA; (d) DCVA; (e) KPCA-Net; (f) DBN; (g) SCCN; (h) CAA; (i) INLPG; and (j) BGAAE.

In addition, Fig. 21 and Table IV show the visualization and quantitative results of the change map, respectively. Fig. 21(j) shows that the result obtained by BGAAE is the closest to the ground truth. Moreover, from Table IV, we can conclude that although the FP index of DCVA and INLPG is 0, their FN index is the highest. On the whole, BGAAE is optimal for the balance of FP and FN, possessing the highest OA, which is

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TABLE IV
RESULTS OF CHANGE DETECTION ON THE BCDD DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
PCA-*k*means	0.8041	3497	4731	0.9634	0.6409
CVA	0.9165	849	4605	0.9758	0.7362
DCVA	0.9445	**0**	6285	0.9721	0.6590
KPCA_Net	0.9683	13313	1463	0.9343	0.5719
DBN	0.8194	83	5677	0.9744	0.6976
SCCN	0.9735	9663	2732	0.9449	0.5889
CAA	0.9832	9563	**1014**	0.9530	0.6652
INLPG	0.9821	**0**	6812	0.9697	0.6208
BGAAE	**0.9955**	1434	1481	**0.9870**	**0.8784**

TABLE V
RESULTS OF ABLATION STUDY ON THE BEIJING DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
Baseline	0.8816	31224	**4116**	0.8586	0.4005
Model-1	0.8465	25649	6731	0.8705	0.3778
Model-2	0.9492	**5985**	6660	**0.9494**	0.6439
Model-3	0.9190	9809	5996	0.9368	0.5979
BGAAE	**0.9507**	7398	5517	0.9483	**0.6572**

TABLE VI
RESULTS OF ABLATION STUDY ON THE NJUST DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
Baseline	0.8149	19790	6718	0.8866	0.2536
Model-1	0.8556	19188	6350	0.8908	0.2767
Model-2	0.9785	8044	1350	0.9598	0.6850
Model-3	0.9839	7117	1247	0.9642	0.7124
BGAAE	**0.9868**	**4534**	**1160**	**0.9756**	**0.7880**

inseparable from semantic-level modeling to exclude the effect from interference factors.

H. Ablation Study

The contributions of each loss term to the overall performance of BGAAE are evaluated through the ablation study.

1) Component Analysis: The proposed bipartite autoencoder with only the reconstruction loss is used as the baseline model. The other two losses and the cycle consistency loss [23] are added one by one to construct the ablation models. These models are as follows: Model-1 is baseline + semantic consistency loss, Model-2 is baseline + domain correlation loss, Model-3 is baseline + domain correlation loss + cycle consistency loss, and BGAAE is baseline + semantic consistency loss + domain correlation loss. Five overall assessment criteria (i.e., AUC, FP, FN, OA, and KC) are used to evaluate the performance of four test VHR image datasets, which are summarized in Tables V–VIII. The change maps are shown in Fig. 22, and Fig. 23 illustrates the comparison of AUCs.

By using the semantic consistency loss, Model-1 can better capture consistent feature representations to effectively exclude the effect from interference factors after the conversion. Accordingly, Model-1 achieved an improvement on the four datasets compared with the baseline model in terms of the AUC, OA, and KC values of Model-1. For instance, on the SZADA dataset, AUC, OA, and KC are improved by

TABLE VII
RESULTS OF ABLATION STUDY ON THE SZADA DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
Baseline	0.9065	20763	3899	0.9014	0.3722
Model-1	0.9373	17989	2414	0.9184	0.4648
Model-2	0.9405	8722	5050	0.9449	0.4979
Model-3	**0.9593**	13535	**1804**	0.9386	0.5576
BGAAE	0.9511	**5671**	2324	**0.9680**	**0.7054**

TABLE VIII
RESULTS OF ABLATION STUDY ON THE BCDD DATASET

Method	*AUC*	*FP*	*FN*	*OA*	*KC*
Baseline	0.9570	37891	**293**	0.8303	0.3342
Model-1	0.9663	26227	814	0.8798	0.4190
Model-2	0.9936	**1210**	1785	0.9866	0.8725
Model-3	0.9699	2374	1306	0.9836	0.8525
BGAAE	**0.9955**	1434	1481	**0.9870**	**0.8784**

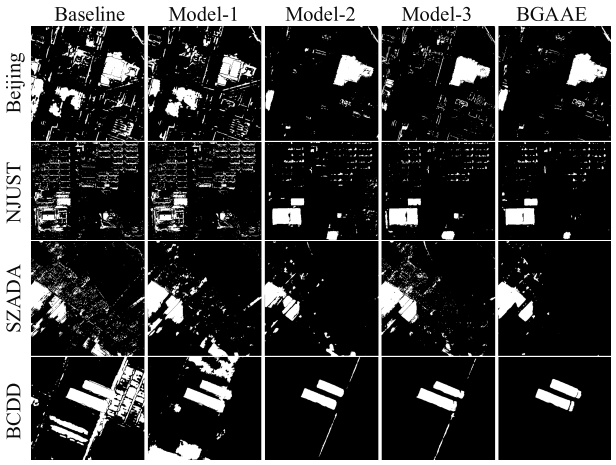


Fig. 22. Change maps obtained by different modules with ablation study on the four datasets.

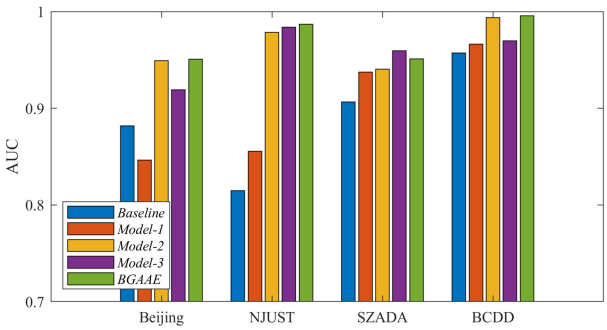


Fig. 23. The AUC of the difference image obtained by ablation study on four datasets.

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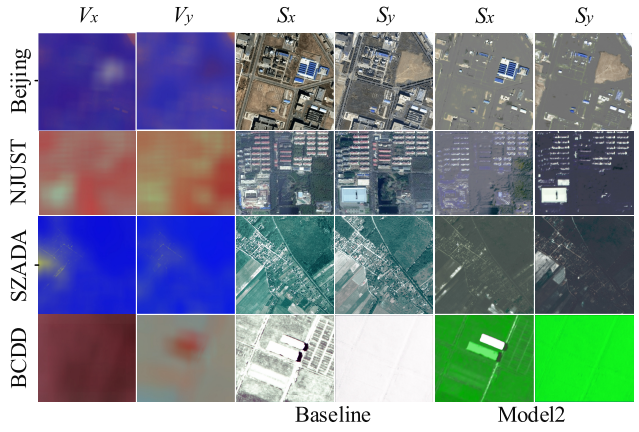


Fig. 24. Effect of the loss terms, organized into one row for each dataset. Col. 1: V_x; Col. 2: V_y; Col. 3: S_x from baseline; Col. 4: S_y from baseline; Col. 5 S_x from model 2; and Col. 6: S_y from model 2.

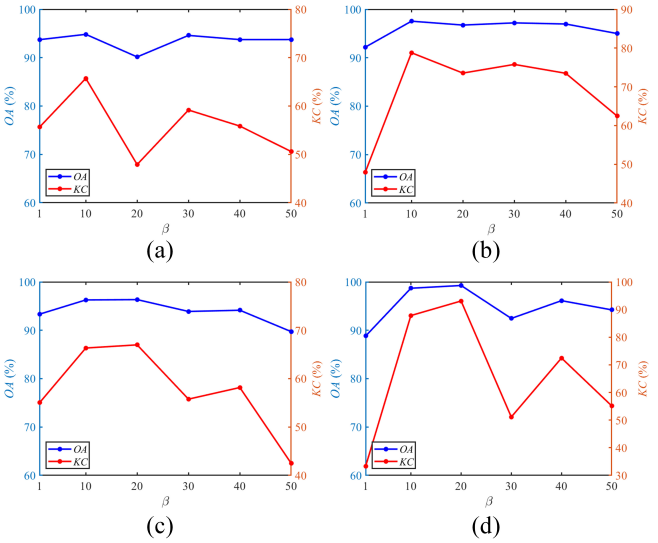


Fig. 25. Relationship between β and the change detection results (OA and KC) on four datasets: (a) Beijing dataset; (b) NJUST dataset; (c) SZADA dataset; and (d) BCDD dataset.

3.08%, 1.7%, and 9.26%, verifying the effectiveness of the semantic consistency loss term.

With the addition of the domain correlation loss, Model-2 dynamically and adaptively aligns the spatial correlation in the latent space, magnifying the changed regions while enforcing the spatial consistency in unchanged regions. To illustrate the performance of the bipartite GAM, Fig. 24 presents visual examples of the representations V_x and V_y. As seen from the first two columns, the bipartite GAM can help to obtain the enhanced feature representations by using spatial information from the opposite images, which explores the consistent regions between the bitemporal images in the latent space domains and highlights the changed regions. Furthermore, the consistent feature representations S_x and S_y obtained with the baseline and model 2 are compared to illustrate the effect of GAM on the domain transformation (the last four columns of Fig. 24). The input images are transformed into the intermediate domain $\mathcal{S}$ in a meaningful way, and the resemblance between the styles of S_x and S_y is clear for Model 2. Accordingly, the qualitative results from Tables V–VIII clearly show the importance of this loss term. There are many fewer noise points on the background regions, and significant improvements were obtained in terms of FP, AUC, OA, and KC for all the test datasets, compared with those of baseline.

Replacing the semantic consistency loss with the cycle consistency loss, Model-3 can be utilized to interpret the ability of the cycle consistency loss compared with Model-2. We can see that the overall performance of Model-3 varies with different datasets. That is, the cycle consistency loss may easily lead to unstable results since it is calculated based on the spectral characteristics in the image domain.

Furthermore, BGAAE contains all components within the DAE framework. It can illustrate the capacity of domain correlation loss compared with Model-1 and the advantage of semantic consistency loss compared with Model-3. The integration of all components enhances the model's ability to resist interference factors and enlarges the difference between the changed regions and background. In addition, this strategy can also enforce spatial consistency.

2) Parameter Analyses: For the proposed total loss function [see (14)], the hyperparameters α and β are used to balance the contributions of L_{recon}, L_{dcl}, and L_{sem}. L_{recon} reproduces faithfully in output, focusing on the reconstruction. L_{sem} aims to maintain a certain degree of semantic consistency for the deep features of the images before and after conversion. L_{dcl} aims to represent the spatial correlation in the latent domain.

Therefore, to conveniently determine the influence of β, we fix α at 1. As β varies by 1, 10, 20, 30, 40, and 50, the performance of BGAAE varies on four different datasets, as depicted in Fig. 25. BGAAE is optimal when $\beta = 10$ for all datasets. When β is less than 10, it dramatically decreases the performance. This indicates the weakened effect of L_{sem}, resulting in the accuracy degradation of the resultant change detection results. A larger β, specifically higher than 10, decreases the effect of domain correlation loss L_{dcl}, resulting in degraded change detection. Thus, the parameter β is set to 10 for all the datasets.

V. Conclusion

This article proposes a novel VHR image change detection method, BGAAE, based on bipartite convolutional autoencoders and GAM. BGAAE is self-supervised and trained using three losses: a reconstruction loss, a domain correlation loss, and a semantic consistency loss. To capture the spatial coupled representations, the domain loss is established at the feature level in the encoder layers. This can effectively construct the spatial correlation and explore for the consistent regions between the bitemporal images, and reduce the influence of change pixels after transcoding on the training objective. In addition, we introduce a semantic consistency loss term to represent the deep semantic information, which is utilized to exclude the impact of different image styles due to interference factors. Based on the DAE framework, this method can avoid the generation of pseudolabels and handle large-scale change

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detection tasks. The comparison analysis with the state-of-the-art models shows outperforming change detection results on four complex VHR datasets. Furthermore, the contribution of each loss function is analyzed through an ablation study.

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